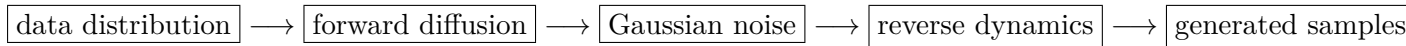


# Mini-project: Generative Diffusion Models from a PDE Perspective

This mini-project introduces generative diffusion models through partial differential equations. The main idea is to view data as a probability distribution, gradually transform this distribution into a simple Gaussian law by a forward diffusion, and then study how to reverse this evolution in order to generate new samples.

The project follows the PDE viewpoint developed by Cao, Johnston, Laurent, Le and Motsch, and then specializes to the Ornstein–Uhlenbeck equation, for which essentially every step can be computed explicitly.

A central objective is to understand the chain



## 1 Generative modeling as evolution of probability densities

Suppose that the available data points

$$X_1, \dots, X_N \in \mathbb{R}^d$$

are sampled from an unknown probability distribution  $\rho_0$ .

A generative model aims to construct new samples whose distribution resembles  $\rho_0$  without simply reproducing the original sample.

Diffusion models proceed in two stages. First, a forward diffusion progressively destroys the structure of  $\rho_0$  and drives it toward a simple reference distribution, usually a Gaussian. Second, a reverse dynamics is learned or approximated and used to transform Gaussian noise back toward the data distribution.

The PDE point of view focuses directly on the evolution of the probability density rather than beginning with neural networks or stochastic differential equations.

## 2 Forward diffusion: the Ornstein–Uhlenbeck equation

Consider the Fokker–Planck equation

$$\partial_t \rho = \nabla \cdot (x \rho) + \Delta \rho, \quad \rho(0, \cdot) = \rho_0. \quad (1)$$

This is the Ornstein–Uhlenbeck equation. Its stationary state is the standard Gaussian density

$$\gamma(x) = (2\pi)^{-d/2} e^{-|x|^2/2}.$$

One of the first goals of the project is to understand why solutions of (1) converge toward  $\gamma$ .

### Associated stochastic process

Equation (1) is the Fokker–Planck equation associated with

$$dX_t = -X_t dt + \sqrt{2} dB_t. \quad (2)$$

The process has the explicit representation

$$X_t = e^{-t} X_0 + \sqrt{1 - e^{-2t}} Z, \quad (3)$$

where  $Z \sim \mathcal{N}(0, I_d)$  is independent of  $X_0$ .

Thus the law of  $X_t$  is obtained by contracting the initial distribution and convolving it with a Gaussian.

### 3 An explicit formula for the density

From (3), the density can be written as

$$\rho_t(x) = \int_{\mathbb{R}^d} \frac{1}{(2\pi\sigma_t^2)^{d/2}} \exp\left(-\frac{|x - e^{-t}y|^2}{2\sigma_t^2}\right) \rho_0(y) dy, \quad (4)$$

where

$$\sigma_t^2 = 1 - e^{-2t}.$$

The student should verify directly that this formula solves (1).

Several simple examples can be studied explicitly:

- $\rho_0 = \delta_{x_0}$ ;
- $\rho_0 = \mathcal{N}(m_0, \Sigma_0)$ ;
- a mixture of two or more Gaussian distributions;
- an empirical distribution  $\rho_0^N = N^{-1} \sum_i \delta_{X_i}$ .

### 4 Why naive time reversal is unstable

Fix a final time  $T > 0$  and define

$$q(s, x) = \rho(T - s, x).$$

A direct change of variables gives

$$\partial_s q = -\nabla \cdot (xq) - \Delta q. \quad (5)$$

The negative diffusion term  $-\Delta q$  makes this equation ill-posed: small high-frequency errors are rapidly amplified.

This is the first major mathematical difficulty in generative diffusion models. One wants to reverse the effect of diffusion without solving an unstable backward heat equation.

### 5 The score and a stable reverse PDE

For a positive density  $\rho$ , define the score

$$s_\rho(x) = \nabla \log \rho(x). \quad (6)$$

The elementary identity

$$\Delta \rho = \nabla \cdot (\rho \nabla \log \rho)$$

allows the backward diffusion term to be rewritten.

Using the known forward solution  $\rho(T - s)$ , one obtains the stable reverse equation

$$\partial_s q = \nabla \cdot ([ -x - 2\nabla \log \rho(T - s, \cdot)] q) + \Delta q. \quad (7)$$

If

$$q(0) = \rho(T),$$

then

$$q(s) = \rho(T - s).$$

The key point is that (7) contains an ordinary positive diffusion term. The missing information needed to reverse the process is encoded in the score  $\nabla \log \rho$ .

## 6 Reverse stochastic dynamics

Equation (7) is the Fokker–Planck equation associated formally with the time-dependent SDE

$$dY_s = [Y_s + 2\nabla \log \rho(T - s, Y_s)] ds + \sqrt{2} dB_s. \quad (8)$$

Starting from a sample close to the Gaussian distribution at time  $T$ , this reverse process transports the law back toward the data distribution.

This is the continuous-time form of the basic mechanism used in score-based generative models.

## 7 The simplest example: one initial data point

Take

$$\rho_0 = \delta_{x_0}.$$

Then

$$\rho_t = \mathcal{N}(e^{-t}x_0, (1 - e^{-2t})I_d).$$

Hence

$$\nabla \log \rho_t(x) = -\frac{x - e^{-t}x_0}{1 - e^{-2t}}. \quad (9)$$

The student can insert (9) into (8). The reverse process is then a linear SDE with time-dependent coefficients and can be analyzed explicitly.

This example is particularly useful pedagogically: the score is known exactly, the forward process is Gaussian, and one can verify directly that the reverse dynamics reconstructs the point  $x_0$  as  $s \rightarrow T$ .

## 8 Gaussian initial data

Suppose now

$$X_0 \sim \mathcal{N}(m_0, \Sigma_0).$$

Then (3) implies

$$X_t \sim \mathcal{N}(m_t, \Sigma_t),$$

with

$$m_t = e^{-t}m_0, \quad \Sigma_t = e^{-2t}\Sigma_0 + (1 - e^{-2t})I_d.$$

The score is therefore

$$\nabla \log \rho_t(x) = -\Sigma_t^{-1}(x - m_t).$$

This gives a second completely explicit reverse model.

Possible exercises are:

- derive the evolution equations for  $m_t$  and  $\Sigma_t$ ;
- check convergence toward  $\mathcal{N}(0, I_d)$ ;
- derive the reverse linear SDE;
- compare simulated moments with the exact formulas.

## 9 Empirical data and Gaussian mixtures

In practice one observes only finitely many samples  $x_1, \dots, x_N$ . The empirical distribution is

$$\rho_0^N = \frac{1}{N} \sum_{i=1}^N \delta_{x_i}.$$

Under the forward Ornstein–Uhlenbeck dynamics,

$$\rho_t^N(x) = \frac{1}{N} \sum_{i=1}^N \mathcal{N}\left(e^{-t}x_i, (1 - e^{-2t})I_d\right)(x). \quad (10)$$

Thus the forward law is an explicit Gaussian mixture.

The student can calculate

$$\nabla \log \rho_t^N(x)$$

explicitly and use it in the reverse dynamics.

This provides a simple toy diffusion model with no neural network at all: the exact score of the noised empirical distribution is available analytically.

## 10 A first numerical generative experiment

A simple one-dimensional or two-dimensional experiment can be organized as follows.

1. Choose a small dataset, for instance two clusters in  $\mathbb{R}^2$ .
2. Apply the forward Ornstein–Uhlenbeck dynamics and visualize  $\rho_t^N$  for increasing times.
3. Compute the exact score of the Gaussian mixture (10).
4. Start particles from an approximately standard Gaussian distribution at time  $T$ .
5. Integrate the reverse SDE numerically.
6. Compare the generated samples with the original dataset.

The student can visualize how Gaussian noise progressively acquires structure during the reverse evolution.

## 11 Why the score must be learned

For a genuine high-dimensional data distribution  $\rho_0$ , the density  $\rho_t$  is not known explicitly. Therefore its score

$$\nabla \log \rho_t$$

is also unknown.

The central learning problem in a score-based diffusion model is thus to approximate the time-dependent vector field

$$(t, x) \mapsto \nabla \log \rho_t(x)$$

from noisy versions of the observed data.

A neural network  $s_\theta(t, x)$  is trained to approximate this score. Once this approximation is available, one replaces the exact score in (8) by  $s_\theta$  and simulates the resulting dynamics.

For an M1 project, this part can remain conceptual. For a stronger M2 student, one can implement a small neural network in one or two dimensions and compare the learned score with the exact score of a Gaussian mixture.

## 12 Denoising interpretation

For the Ornstein–Uhlenbeck process,

$$X_t = e^{-t} X_0 + \sigma_t Z.$$

Thus observing  $X_t$  amounts to observing a noisy version of  $X_0$ .

The score contains information about how to move a noisy point toward regions where the original data are likely to lie. This gives the reverse drift a natural denoising interpretation.

The student can explore numerically the vector field

$$x \mapsto \nabla \log \rho_t(x)$$

for a mixture of Gaussians and observe how it points toward high-density regions.

## 13 Optional extension: deterministic probability flow

The reverse evolution can also be represented by a deterministic transport equation. In the present normalization, one may consider

$$\partial_s q = \nabla \cdot ([ -x - \nabla \log \rho(T - s, \cdot) ] q). \quad (11)$$

This equation has no diffusion term and corresponds to a deterministic flow of particles.

The student can compare the stochastic reverse dynamics (8) and the deterministic probability-flow dynamics (11).

## 14 Optional extension: overfitting and generalization

The PDE viewpoint also raises an interesting conceptual issue. If the initial distribution is exactly the empirical measure

$$\rho_0^N = \frac{1}{N} \sum_i \delta_{x_i},$$

and if the score and reverse dynamics are computed exactly, then the reverse process reconstructs the empirical distribution itself.

Thus a mathematically exact reverse process need not create genuinely new information: it can simply return the training data.

This gives a natural discussion topic for a stronger student:

- What is meant by generalization in a generative model?
- Why can imperfect score estimation act as a form of regularization?
- How does finite-time stopping influence the generated distribution?

## 15 Suggested project organization

### Part 1. Forward diffusion.

Study the Ornstein–Uhlenbeck PDE and SDE and derive the explicit solution.

### Part 2. Gaussian examples.

Compute the evolution and score for a Dirac mass and for Gaussian initial data.

### Part 3. Reverse PDE.

Derive the stable reverse equation from the forward Fokker–Planck equation.

### Part 4. Numerical generation.

Simulate forward and reverse dynamics for a simple one- or two-dimensional dataset.

### Part 5. One extension.

Choose one among:

- exact score for an empirical Gaussian mixture;
- score learning with a small neural network;
- stochastic versus deterministic reverse dynamics;
- convergence of the forward Ornstein–Uhlenbeck semigroup;
- overfitting and generalization for empirical data.

## 16 Suggested scope

For an **M1 project**, a reasonable objective is:

- understand diffusion models as evolution of probability densities;
- derive and solve the Ornstein–Uhlenbeck forward equation;
- understand the score  $\nabla \log \rho_t$ ;
- derive the reverse PDE formally;
- implement forward and reverse simulations in dimension one or two.

For a stronger **M2 project**, one could additionally study:

- score matching and denoising score matching;
- approximation of the score by a neural network;
- stability with respect to errors in the score;
- entropy or Wasserstein convergence of the forward Ornstein–Uhlenbeck equation;
- the relation between stochastic reverse diffusion and deterministic probability flow;
- the question of overfitting versus generalization.

## 17 Core references

### References

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- [3] J. Ho, A. Jain and P. Abbeel, *Denoising diffusion probabilistic models*, Advances in Neural Information Processing Systems, 33 (2020).
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